

Product Requirement Document (PRD)

Software Requirements Specification for Real-Time Audio-Guided Recipe Execution App

Project:	Recipe Execution Assistant	Version:	1.0 (v1 Release)
Document:	Functional & System SRS	Date:	August 10, 2026

1. Executive Summary & Core Unique Selling Proposition (USP)

Core Value Proposition

Standard recipe platforms display ingredients as a static upfront list and bury instructions inside block text. This app functions as a **Live Execution Assistant & Kitchen Co-Pilot**. It shifts the cooking experience from a reference card to a real-time hands-free timeline by integrating step-specific durations, dynamic flame/heat settings, contextual ingredient substitutes, and interactive multi-voice guidance into a unified flow.

2. General Principles & Core System Logic

- Flame/Heat Guidance Scope:** Heat instructions apply strictly to the flame/heat intensity of the cooking vessel (e.g., pan, pot) when adding or simmering ingredients. Ingredients added without active vessel heating (e.g., garnishes, dry seasonings added without active vessel heating) omit heat instructions entirely.
- Option Item Timing:** Optional ingredients inherit default step timing without altering overall recipe timeline duration.
- Non-Intrusive Alternative Selection:** Users are never forced to respond to interactive UI prompts or popups to select alternative ingredients. Alternatives are announced verbally, allowing the user to select and execute choices manually.
- Timer Persistence:** Step countdown timers run continuously in the background without pausing during voice announcements, audio staggering, or overlay interactions.

3. Screen Views & Specifications

A. Main View: Single Scrollable Recipe Timeline

A continuous linear horizontal-row list displaying all recipe steps line-by-line for quick scanning prior to or during cooking.

Column Name	Visual Specification & Behavior
1. Step	Automatically calculated sequential step number (1, 2, 3...).
2. Flame level	Displays only the color-coded stove flame icon (🔴 High, 🔸 Medium, 🔵 Low, or blank if no vessel heat) without text descriptions.
3. Duration	Specific timer duration for the step (formatted as MM:SS or HH:MM:SS).

Column Name	Visual Specification & Behavior
4. Time	Accumulative elapsed recipe timestamp up to that step.
5. Ingredient & Quantity	Multi-line text field where each item is placed on a new line for maximum readability. Includes both explicit text and corresponding emoji icons (3 🍵, 1/2 🥄, 🍴, 🌱, 🌶️).
6. Alternatives	Plain text field displaying alternative ingredient options (e.g., Carrot & Beans or Raw Mango). Left completely blank if no alternatives exist.

Timeline Authoring & Interaction Rules

- **Add Step Button:** Positioned explicitly at the bottom of the timeline (+ Add Step) to append a new empty row without cluttering the screen.
- **Step Reordering:** Drag handles (::) allow reordering steps, which automatically recalculates step numbers, accumulative times, and interactive execution sequences.
- **Step Deleting:** Clicking the trash icon (🗑️) immediately deletes a step and presents a temporary **"Undo"** toast banner at the bottom.
- **Interactive Mode Lock:** Timeline reordering and editing are strictly **locked** while Interactive Mode is actively running.

B. Interactive Cooking Mode (Audio & UI Co-Pilot)

Triggered when tapping *"Start Interactive Mode"*, transforming the UI into a real-time hands-free co-pilot.

1. On-Screen Control & Multitasking Layout

Layout **Dynamic Side-by-Side Cards:** Supports displaying up to **2 or 3 active concurrent processes** side-by-side in responsive cards with independent Play/Pause, Step Forward/Back buttons, and countdown bars.

Overflow **Compact Numbered Circular Badges (4+ Tasks):** If 4 or more concurrent tasks run simultaneously, active cards are capped at 3, and additional tasks render as compact numbered circular icons. Tapping an icon opens a detail drawer overlay.

UX Logic **Uninterrupted Overlay Experience:** Opening or interacting with a detail overlay does **not** interrupt or pause active background timers or voice audio streams.

2. Audio & Voice Guidance Logic

- **Multi-Voice Profiles:** Primary cooking steps utilize a primary voice profile (e.g., Male Voice). Concurrent background or parallel prep tasks utilize distinct secondary voice profiles (e.g., Female Voice).
- **Sequential Audio Staggering:**
 - No two audio prompts play simultaneously. Overlapping prompts are scheduled a few seconds apart.
 - Background timers continue running silently while awaiting their audio slot.
 - Delayed audio prompts announce the direct action prompt (e.g., "*Stir onions now*") without dynamic time recalculation.
- **Voice Prompt Structure & Conditional Rules:**
 - **Heat Shift Rule:** Read heat level *before* ingredient name (e.g., "*Set heat to low, then add onions*"). If the next step requires the exact same heat, omit the heat instruction.
 - **Alternatives Announcement:** Read main preference first, followed by options: "*Set heat to low, then add Tomato... or alternatively, Mango*".
 - **Optional Items Announcement:** Precede optional items with explicit phrasing: "*Optionally, set heat to low and add cilantro*".
 - **Timer Autostart:** Step countdown timer launches automatically as soon as the voice assistant finishes reading the step instructions.
- **Resting & Completion Phases:**
 - Resting/cooling steps (e.g., 5 min rest) use the primary voice profile and standard timer card UI.
 - Upon reaching 00:00 on the final step, the assistant announces: "*Cooking complete! Enjoy your meal*", plays a completion chime, and automatically transitions back to the main scrollable timeline.